Brian N. White

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EDUCATION

Ph.D., Statistics & Operations Research May 2026

University of North Carolina at Chapel Hill

Dissertation: *Scalable Spatial Extreme Value Data Fusion with Applications to Coastal Hazard Projection Under a Changing Climate.* Advisor: Richard L. Smith.

M.A., Mathematical Statistics 2020

Wake Forest University · Full scholarship with stipend

B.A., Environmental Studies (Ecology) 2018

University of North Carolina at Asheville

RESEARCH INTERESTS

Spatial extreme-value theory, Bayesian hierarchical modeling, scalable Gaussian processes, spatial data fusion, environmental statistics and public health, spatial epidemiology

EXPERIENCE

Biostatistician 2022 – Present

Wake Forest University School of Medicine, Dept. of Biostatistics & Data Science

Opioid misuse surveillance: implementing and contributing to the development of Bayesian models that integrate fragmented data sources across spatial and temporal scales to estimate the size of hidden populations, and working with state health departments to turn these estimates into operational tools. Supported by NIDA R01DA052214 (MPIs: Kline, Hepler). NIH LEAP trial: contributing statistician for anti-obesity medication RCT. Clinical collaboration across surgery, pediatrics, gerontology, and gastroenterology. Causal inference on observational EHR data.

Graduate Research Assistant 2020 – 2026

University of North Carolina at Chapel Hill, Dept. of Statistics & Operations Research

Forecasting extreme coastal water levels across climate scenarios by fusing physics-based ocean simulations with historical tide gauge observations via scalable Gaussian processes.

PUBLICATIONS

1. Wylie, Leonard, **White**, et al. Perioperative chemotherapy for clinical stage IIIB gallbladder cancer: A review of the National Cancer Database. *International Journal of Clinical Oncology*, 2026.
2. Cyrille-Superville, Patel, **White**, et al. Sex-specific trends in temporary mechanical circulatory support use as bridge to orthotopic heart transplant. *JACC: Advances*, 2026.
3. Young, Rives, Gudzune, Jaeger, Simmons, **White**, et al. The long-term effectiveness of the anti-obesity medication phentermine (LEAP) trial: Rationale, design, and baseline characteristics. *Contemporary Clinical Trials*, 2026.
4. Jeong, Edmunds, McCollum, **White**, Clayton. Lower mean nocturnal baseline impedance in erosive reflux disease, Barrett's Esophagus, and elevated body mass index compared to nonerosive reflux disease. *Diseases of the Esophagus*, 2026.
5. Kline, **White**, Lancaster, et al. Estimating prevalence of opioid misuse in North Carolina counties from 2016–2021: An integrated abundance model approach. *Epidemiology*, 2025.
6. Lippert, Callahan, **White**, et al. Hospitalist and "SNFist" clinician perspectives on frailty assessment of rehab patients. *Journal of General Internal Medicine*, 2025.
7. Semelka, **White**, Callahan, Pajewski. The association of frailty with post-hospital discharge location and health outcomes. *Journal of the American Geriatrics Society*, 2025.
8. Heeren, Ruddiman, Simmons, **White**, et al. Application of the Lancet Commission criteria for the diagnosis of obesity to a clinical trials population: The LEAP trial. *Obesity*, 2025.
9. Krakovski, Madigan, Cecil, **White**, et al. Patient factors contributing to wireless pH monitoring intolerance. *Journal of Clinical Gastroenterology*, 2025.
10. Brown, **White**, Hanchate, et al. Short term health outcomes of the Weaver fertilizer plant fire on the surrounding community. *Journal of Public Health Management & Practice*, 2025.
11. Estadt, **White**, Ricks, et al. The impact of fentanyl on state- and county-level psychostimulant and cocaine overdose death rates by race in Ohio from 2010 to 2020: A time series and spatiotemporal analysis. *Harm Reduction Journal*, 2024.
12. Khanna, Banga, Rigdon, **White**, et al. Role of continuous pulse oximetry and capnography monitoring in the prevention of

postoperative respiratory failure, postoperative opioid-induced respiratory depression and adverse outcomes on hospital wards: A systematic review and meta-analysis. *Journal of Clinical Anesthesia*, 2024.

13. Kipp, Vesely, Lance, **White**, et al. Age influence on total ankle arthroplasty outcomes: A systematic review. *Journal of Foot and Ankle Surgery*, 2024.
14. Vesely, Kipp, Lance, **White**, et al. BMI influence on total ankle arthroplasty outcomes: A systematic review. *Foot & Ankle Surgery: Techniques, Reports & Cases*, 2024.
15. Fabian, Adkins, **White**, et al. Temporal trends in BAHA softband wear time among pediatric patients. *International Journal of Pediatric Otorhinolaryngology*, 2024.

MANUSCRIPTS

White, Blanton, Luettich, Smith. Fusing sparse observations and dense simulations for spatial extreme value analysis: Application to U.S. coastal sea levels. *arXiv preprint*, 2026. arXiv · code

Lassiter, **White**, et al. Parental concerns about tap water associated with 4–6 year-old children's increased consumption of sugar-sweetened beverages. In revision.

PRESENTATIONS

Presenting author underlined.

Laks, Gill, Woolworth, **White**, Seibert. Impact of maternal BMI on primary cesarean delivery following labor induction: A population-based study. Carolinas Medical Center Department of Ob/Gyn Resident Research Symposium, 2026. Oral presentation (first place).

Mayfield, Gill, **White**, Seibert. Unseen influences: Implicit bias and maternal mortality across the U.S. Carolinas Medical Center Department of Ob/Gyn Resident Research Symposium, 2026. Oral presentation (second place).

Laks, Gill, Woolworth, **White**, Seibert. Impact of maternal BMI on primary cesarean delivery following labor induction. North Carolina Obstetrical and Gynecological Society, 2026. Poster.

White, Hepler, Hu, Jetson, Xiong, Smith, Delcher, Kline. Estimation of the prevalence of opioid misuse in Washington State counties, 2017–2023: A Bayesian spatiotemporal abundance model approach. Wake Forest Public Health Sciences Staff Science Day, 2025. Poster; one of five selected from ∼100 to give a division-wide Ignite talk.

White, Blanton, Luettich, Smith. Combining observational and model data for spatial extremes through a multivariate Gaussian latent process. 14th International Conference on Extreme Value Analysis (EVA 2025). Poster.

TEACHING

Led course sections, University of North Carolina at Chapel Hill

Discrete Mathematics, weekly seminar (Spring 2025): gave 50-minute lectures reviewing the week's proof concepts, with self-prepared materials and problem-solving drills.

Machine Learning, Python computing lab (Spring 2026).

Guest lectures, University of North Carolina at Chapel Hill

Introductory Statistics: Bernoulli random variables and the distributions derived from them (binomial, geometric, etc.).

Machine Learning: predictive modeling with tidymodels (R).

Teaching assistant

UNC Chapel Hill & Wake Forest University (2018–2026): Machine Learning, Statistical Modeling, Biostatistics, Data Analysis, Discrete Mathematics, and Regression; office hours, grading, proctoring, and gradebook management.

Wake Forest University School of Medicine: introductory statistics and statistical models for physicians and medical students.

SERVICE

Search committee member, Staff Biostatistician search, Dept. of Biostatistics & Data Science, Wake Forest University School of Medicine, 2023; ongoing participation in candidate interviews.

Capstone committee member for Jennifer Wylie, M.D. (M.S. in Translational and Health System Science), Wake Forest University School of Medicine, 2026.

SKILLS

Foundations: Measure-theoretic probability, asymptotic theory, stochastic processes, optimization, real and functional analysis

Methods: Bayesian hierarchical modeling, MCMC, spatial/spatiotemporal statistics, extreme-value theory, Gaussian processes, causal inference, experimental design, survival analysis, mixed models

Computing: R, Python, Stan, JAGS, SQL, LaTeX, Git, Linux, HPC/Slurm, parallel computing, reproducible workflows (targets, renv)

PROFESSIONAL MEMBERSHIPS

Member, American Statistical Association.

Classical piano background (Shepherd School of Music, Rice). Student of the Taubman approach.